

Magnesium-25

²⁵Mg (Mg I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 18 levels, 61 lines.

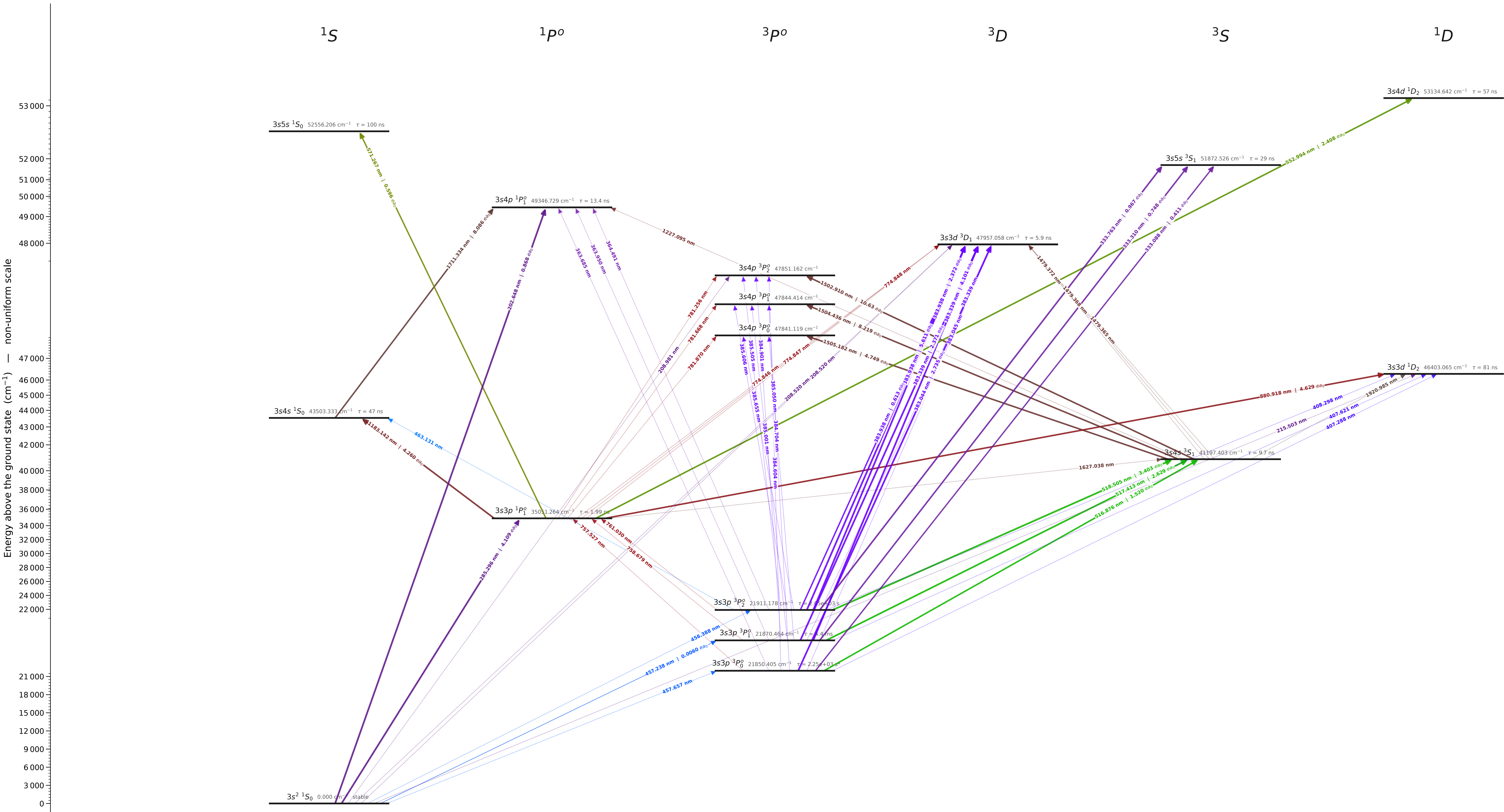
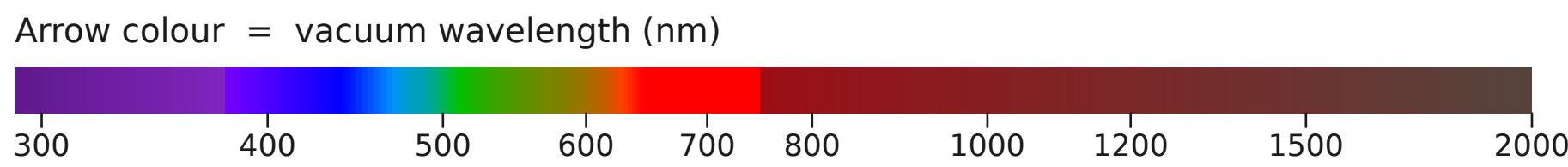
Mg⁺ ionisation limit 61671.05 cm⁻¹ = 7.64624 eV (162.151 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U \mathbf{e} U' \rangle $ (ea0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3s^2\ ^1S_0 - 3s3p\ ^1P_1$	285.2964	285.2126	1050.8105	4.109	4.910e+08	79.98	97.7	singlet cooling / MOT / imaging line
$3s3p\ ^3P_0 - 3s3d\ ^3D_1$	383.0441	382.9355	782.6578	2.735	8.990e+07	26.98	53	triplet MOT / repumping
$3s3p\ ^3P_0 - 3s3d\ ^3D_2$	383.0446	382.9359	782.6568	-	2.110e-05	26.98	0	triplet MOT / repumping
$3s3p\ ^3P_1 - 3s3d\ ^3D_1$	383.3387	383.2299	782.0564	2.371	6.740e+07	26.98	39.8	triplet MOT / repumping
$3s3p\ ^3P_1 - 3s3d\ ^3D_2$	383.3389	383.2301	782.0560	-	2.200e-05	26.98	0	triplet MOT / repumping
$3s3p\ ^3P_2 - 3s3d\ ^3D_2$	383.3391	383.2304	782.0555	4.101	1.210e+08	26.98	71.4	triplet MOT / repumping
$3s3p\ ^3P_2 - 3s3d\ ^3D_1$	383.9379	383.8290	780.8358	0.613	4.480e+06	26.98	2.64	triplet MOT / repumping
$3s3p\ ^3P_2 - 3s3d\ ^3D_2$	383.9381	383.8292	780.8355	5.611	1.610e+08	26.98	95	triplet MOT / repumping
$3s3p\ ^3P_2 - 3s3d\ ^3D_2$	383.9383	383.8295	780.8349	2.372	4.030e+07	26.98	23.8	triplet MOT / repumping
$3s^2\ ^1S_0 - 3s3p\ ^3P_1$	457.2377	457.1096	655.6600	0.006	2.540e+02	3.617e-05	100	intercombination line: narrow-line spectroscopy / Ramsey...
$3s^2\ ^1S_0 - 3s3p\ ^3P_0$	457.6574	457.5292	655.0587	-	5.800e-04	7.074e-11	100	optical lattice clock transition
$3s3p\ ^3P_0 - 3s4s\ ^3S_1$	516.8761	516.7321	580.0084	1.520	1.130e+07	16.41	11.2	triplet optical pumping / repumping, metastable-beam spec...
$3s3p\ ^3P_1 - 3s4s\ ^3S_1$	517.4125	517.2684	579.4071	2.629	3.370e+07	16.41	33.9	triplet optical pumping / repumping, metastable-beam spec...
$3s3p\ ^3P_2 - 3s4s\ ^3S_1$	518.5048	518.3604	578.1865	3.403	5.610e+07	16.41	54.9	triplet optical pumping / repumping, metastable-beam spec...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

^{25}Mg hyperfine structure ($I = 5/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)					
3s3p $^3P_0^o$	-144.977(5)	-8.308(5)	F=3/2: +501.60	F=5/2: +151.62	F=7/2: -364.52			
3s3p $^3P_2^o$	-128.445(5)	16.009(5)	F=1/2: +910.32	F=3/2: +710.45	F=5/2: +381.33	F=7/2: -71.03	F=9/2: -638.22	
3s4s 3S_1	-321.6(15)	-	F=3/2: +1125.60	F=5/2: +321.60	F=7/2: -804.00			

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3s^2\ ^1S_0 - 3s4p\ ^1P_1^o$	202.648	26-24	2918.9(16)
$3s^2\ ^1S_0 - 3s4p\ ^1P_1^o$	202.648	25-24	1526.4(16)
$3s^2\ ^1S_0 - 3s3p\ ^1P_1^o$	285.296	26-24	1414(8)
$3s^2\ ^1S_0 - 3s3p\ ^1P_1^o$	285.296	25-24	741(6)
$3s3p\ ^3P_0^o - 3s3d\ ^3D_1$	383.044	26-24	60(15)
$3s3p\ ^3P_0^o - 3s3d\ ^3D_2$	383.045	26-24	60(15)
$3s3p\ ^3P_1^o - 3s3d\ ^3D_1$	383.339	26-24	61(3)
$3s3p\ ^3P_1^o - 3s3d\ ^3D_3$	383.339	26-24	61(3)
$3s3p\ ^3P_2^o - 3s3d\ ^3D_2$	383.339	26-24	61(3)
$3s3p\ ^3P_2^o - 3s3d\ ^3D_3$	383.938	26-24	58(4)
$3s3p\ ^3P_2^o - 3s3d\ ^3D_3$	383.938	26-24	58(4)
$3s3p\ ^3P_2^o - 3s3d\ ^3D_2$	383.938	26-24	58(4)
$3s^2\ ^1S_0 - 3s3p\ ^3P_1^o$	457.238	26-24	2683.180(20)
$3s^2\ ^1S_0 - 3s3p\ ^3P_1^o$	457.238	25-24	1405.20(10)
$3s^2\ ^1S_0 - 3s3p\ ^3P_0^o$	457.657	26-24	2681.467(20)
$3s^2\ ^1S_0 - 3s3p\ ^3P_0^o$	457.657	25-24	1403.90(10)
$3s3p\ ^3P_0^o - 3s4s\ ^3S_1$	516.876	26-24	-391.3(17)
$3s3p\ ^3P_0^o - 3s4s\ ^3S_1$	516.876	25-24	-205.7(15)
$3s3p\ ^3P_1^o - 3s4s\ ^3S_1$	517.413	26-24	-390.1(14)
$3s3p\ ^3P_1^o - 3s4s\ ^3S_1$	517.413	25-24	-209.1(13)
$3s3p\ ^3P_2^o - 3s4s\ ^3S_1$	518.505	26-24	-394.4(8)
$3s3p\ ^3P_2^o - 3s4s\ ^3S_1$	518.505	25-24	-205.7(8)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$